

Polarity Bench

Separating a polar bond from a polar molecule, and finding the number that turns one into the other.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/polarity-bench/

1 Before you open anything

Answer from what you already know. Getting it right is not the point — having it on paper, in ink, so you can compare later, is.

PREDICTION

CO₂ contains two C=O bonds. Oxygen ($\chi = 3.44$) pulls harder on the shared electrons than carbon ($\chi = 2.55$), so each of those bonds is polar. Now suppose you could raise oxygen's electronegativity to 4.00, the top of the Pauling scale, without changing the molecule's shape. What happens to the *net dipole of the whole molecule*: does it grow, shrink, or stay where it is? Give a number if you can.

Write one or two sentences saying what reasoning got you there.

2 Reading the bench

Open the demonstration. It starts on CO₂ with every electronegativity at its real Pauling value. Two panels do the reporting: *Every bond in this molecule* describes the bonds one at a time, and *The whole molecule* describes what they add up to.

1. Confirm you are on CO₂. If any slider has been moved, press **Real values**.
2. From *Every bond in this molecule*, record the C–O bond's $\Delta\chi$ and its % ionic character.
3. From *The whole molecule*, record **Net molecular dipole**, **Shape** with its bond angle, **Lone pairs on the central atom**, and **Measured dipole moment of the real molecule**.
4. Copy the badge printed beside the molecule's name at the top of the stage into the last column.

5. Press **H₂O** in the *Molecule* panel and fill in the second row the same way.

Molecule	Bond $\Delta\chi$	% ionic character	Shape (bond angle)	Lone pairs on centre	Net molecular dipole ($\Delta\chi$ units)	Measured μ (D)	Badge
CO ₂							
H ₂ O							

a. Both molecules have exactly two bonds, and in both molecules every bond is polar. Name the two columns in your table that are *not* the same for the two molecules.

b. Which of the two has the more polar *bond*? Which has the larger net dipole as a *molecule*? Answer both, using your numbers.

c. Look at the second panel, *The vector sum*, while CO₂ is selected. Describe in one sentence what the chain of arrows does after the last arrow is laid down.

The moment you drag any electronegativity slider, the *Measured dipole moment* box changes to a dash. That is deliberate: measured values belong to real molecules, and once you have invented an electronegativity there is no experiment to quote. Everything else on the page is still computed honestly from the sliders.

3 What actually controls the net dipole

One slider, moved through a range, on two molecules that differ only in shape. Keep the central atom at its real value throughout; move only the outer-atom slider.

1. Select **CO₂** and press **Real values**. Leave **χ of C (central atom)** at 2.55 for the whole of this table.
2. Set **χ of O (both outer atoms)** to each value in the first column and record the bond $\Delta\chi$ and the net dipole. Then compute the last column yourself.

$\text{CO}_2 - \chi \text{ of O}$	Bond $\Delta\chi$	Net molecular dipole	Net $\div \Delta\chi$
2.80			
3.44 (real)			
4.00			

- Now press **H₂O** and **Real values**. Leave **χ of O (central atom)** at 3.44 throughout.
- Set **χ of H (both outer atoms)** to each value below and fill the row in. Give the last column to two decimal places.

$\text{H}_2\text{O} - \chi \text{ of H}$	Bond $\Delta\chi$	Net molecular dipole	Net $\div \Delta\chi$
2.20 (real)			
1.70			
1.20			
0.80			

- a.** The last column is nearly constant down each table, and the two tables give different constants. Write both constants.

- b.** Complete the rule in your own words: the net molecular dipole equals the bond $\Delta\chi$ multiplied by a factor that depends only on _____, and for CO_2 that factor is _____.

- c.** The bond angle in water is 104.5° , so each bond dipole makes an angle of 52.25° with the molecule's axis of symmetry. Evaluate $2 \cos(52.25^\circ)$. Compare it with the constant you found for H_2O .

d. Go back to your prediction in Part 1. Was it right? If it was not, state precisely which assumption failed — not “I guessed”, but the specific thing you believed about how bond polarity and molecular polarity are connected.

4 Working backwards, and a shortcut that breaks

Backwards. A test question rarely hands you the sliders in the order you want them.

1. Stay on **H₂O**. Using the constant from Part 3, calculate on paper the $\Delta\chi$ that would give a net molecular dipole of exactly **2.00** $\Delta\chi$ units.
2. From that $\Delta\chi$, and with χ of O held at 3.44, calculate the χ of H you would need, to two decimal places.
3. Set the χ of H slider there and record what *Net molecular dipole* actually reads.

a. Required $\Delta\chi$ = _____ χ of H = _____ Reading obtained = _____

If the reading is not exactly 2.00, say what limits how close you can get.

The shortcut. "More polar bonds means a more polar molecule" is a reasonable-sounding rule. This pair takes it apart while holding the shape fixed.

1. Press **CCl₄**, then **Real values**. Record the row below.
2. Move the **Rotate the molecule** slider to **45**, then **120**, then **-90**, watching the net dipole readout.
3. Press **CHCl₃**, then **Real values**, and record its row. It has two kinds of bond, so the bond panel lists two.
4. With CHCl₃ still selected, press **Make outer atoms identical** and record the last row.

Setting	Shape (bond angle)	Bonds and their $\Delta\chi$	Net molecular dipole	Badge
CCl ₄ , real values				
CHCl ₃ , real values				
CHCl ₃ , outer atoms made identical				

b. Which of the two real molecules has more polar bonds? Which one is the polar molecule? State what this does to the shortcut.

c. At the three rotation angles you tried on CCl_4 , the drawing changed but the *Net molecular dipole* readout did not. Why must that be so for a vector sum that is genuinely zero?

d. In the last row you changed one electronegativity and nothing geometric. Record the χ of H the button set, and explain in one sentence why that single change sends the net dipole to 0.00.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it and check yourself; where you differ, find the step that went wrong rather than just correcting the answer. Pauling values you may need: H 2.20, B 2.04, C 2.55, N 3.04, O 3.44, F 3.98, Cl 3.16.

1. A bent AB_2 molecule has a bond angle of 104.5° and each A–B bond has $\Delta\chi = 1.90$. Predict its net molecular dipole in $\Delta\chi$ units. Then state what χ of H you would set on H_2O to reproduce this case in the demonstration.

2. The four C–Cl bond dipoles of CCl_4 sum to exactly zero. Use only that fact and the χ values above — no trigonometry — to find the net molecular dipole of CHCl_3 in $\Delta\chi$ units, and state which end of the molecule the net arrow points towards.

3. Compute $\Delta\chi$ for a B–F bond, then its percent ionic character from Pauling's relation, % ionic = $100 (1 - e^{-0.25 \Delta\chi^2})$. BF_3 has three of these bonds – the most ionic bonds anywhere in this demonstration – and a net dipole of zero. Explain how both statements can be true at once.

4. The panel *What this model leaves out* reports that the model comes out 16% below the measured dipole for H_2O and 35% below it for NH_3 , while its predictions of *zero* for CO_2 , CCl_4 and BF_3 are exact and need no calibration. Explain both halves of that: why the model undershoots for those two molecules in particular, and why the zeros are safe from the same error. No calculation required.

6 On your own

Nothing here is collected. The demonstration holds more than this worksheet used.

- Leave *Show $\delta+$ / $\delta-$ on the atoms* switched on and look at CO_2 . The atoms carry partial charges while the molecule is labelled nonpolar. Settle for yourself what “nonpolar molecule” is and is not claiming.
- Select HF and find where its marker lands on the $\Delta\chi$ scale relative to the 1.7 line. HF is a covalent gas. Decide what the 1.7 line is actually good for.
- Compare BF_3 and NH_3 : three bonds each, one flat and one pyramidal. Use *Rotate the molecule* to convince yourself which is which, then predict the net dipole of each before you read it.
- In *All seven at their real values*, read the bond $\Delta\chi$ column against the last column and find every molecule that has polar bonds and is nonpolar anyway.